

CARTERSVILLE, GA, USA

WMO: 722156

Lat: 34.123N

Lon: 84.849W

Elev: 763

StdP: 14.30

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53873

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.5	(c) 23.4	(d) 5.6	(e) 7.6	(f) 27.0	(g) 9.5	(h) 9.2	(i) 30.2	(j) 15.3	(k) 46.6	(l) 12.8	(m) 46.2	(n) 2.4	(o) 340	(p) 0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	94.1	74.0	91.9	74.0	90.0	73.8	77.9	87.4	77.0	86.5	76.1	85.2	4.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	136.3	81.2	74.3	131.8	80.4	73.3	127.4	79.5	41.7	87.3	40.7	86.7	40.0	85.3	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.5	(b) 10.9	(c) 9.7	(e) 13.5	(f) 97.1	(g) 5.0	(h) 3.4	(i) 9.9	(j) 99.5	(k) 7.0	(l) 101.5	(m) 4.2	(n) 103.4	(o) 0.6	(p) 105.9		
(a) 12.5	(b) 10.9	(c) 9.7	(e) 11.9	(f) 80.0	(g) 4.7	(h) 1.3	(i) 8.5	(j) 80.9	(k) 5.7	(l) 81.7	(m) 3.0	(n) 82.4	(o) -0.4	(p) 83.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.3	41.8	45.3	53.0	60.9	68.7	76.1	78.5	78.6	73.1	62.1	51.4	45.2
	DBStd	14.94	9.95	8.82	8.61	6.94	6.30	3.85	3.13	3.56	5.44	7.93	8.16	8.78
	HDD50	846	294	181	70	8	0	0	0	0	0	8	77	208
	HDD65	3033	720	552	379	157	41	0	0	0	8	148	412	616
	CDD50	4970	39	51	162	334	579	784	882	886	692	383	120	58
	CDD65	1682	0	1	7	33	155	334	417	421	250	58	5	1
	CDH74	15073	1	7	95	448	1463	3044	3663	3757	2087	477	29	2
	CDH80	6144	0	0	8	73	465	1322	1605	1722	843	104	2	0
Wind	WSAvg	2.9	3.2	3.5	3.7	3.5	2.8	2.2	1.9	2.2	2.6	2.7	2.9	3.1
Precipitation	PrecAvg	50.5	4.8	4.6	5.6	3.9	3.8	3.9	4.8	3.8	3.8	3.3	4.1	4.1
	PrecMax	68.7	7.8	9.1	9.6	7.6	10.2	6.7	10.7	8.7	10.9	9.0	7.9	8.4
	PrecMin	28.0	2.3	1.9	2.0	1.7	1.0	1.6	0.7	1.3	0.4	0.3	1.6	1.3
	PrecStd	8.7	1.5	1.6	2.2	1.4	2.0	1.6	2.2	1.7	2.7	2.2	1.7	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.2	74.2	81.0	85.5	91.1	96.0	95.1	98.0	95.5	87.0	78.5	72.5
		MCWB	60.6	63.7	63.6	66.2	71.1	73.0	74.8	74.3	72.2	70.5	65.3	63.4
	2%	DB	66.4	68.8	76.6	82.1	87.7	92.6	92.8	94.1	91.4	82.4	73.0	68.3
		MCWB	58.2	59.3	61.7	64.9	70.0	73.2	75.4	74.6	71.3	67.8	62.0	62.3
	5%	DB	62.8	65.5	72.9	79.5	85.1	90.3	90.6	91.3	88.4	79.4	70.0	64.2
		MCWB	56.5	57.2	59.5	63.5	69.0	73.2	75.5	74.4	71.4	65.8	60.4	58.5
	10%	DB	58.5	61.4	68.8	76.0	82.3	87.7	88.4	89.1	85.0	75.6	66.2	60.1
		MCWB	53.5	52.5	57.1	62.0	68.6	72.6	75.0	74.1	71.0	64.5	58.6	55.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.7	65.4	66.1	70.2	74.7	78.1	79.5	78.9	76.2	74.5	70.1	67.1
		MCDB	68.0	71.2	75.8	80.2	82.4	87.2	88.1	88.8	85.3	81.6	72.7	69.7
	2%	WB	60.7	62.8	64.2	68.1	73.0	76.8	78.1	77.6	75.0	71.8	66.0	63.3
		MCDB	64.6	67.3	72.7	77.6	82.3	86.4	87.3	87.7	84.3	78.5	69.8	66.7
	5%	WB	57.9	59.2	62.2	66.2	71.5	75.7	77.2	76.7	73.8	69.7	63.2	59.9
		MCDB	62.0	63.4	69.6	75.1	81.0	85.1	86.5	86.4	82.6	75.1	67.2	63.3
	10%	WB	54.0	55.2	59.7	64.3	70.4	74.6	76.3	75.7	72.6	67.3	60.0	56.0
		MCDB	57.6	59.5	66.4	72.4	79.2	83.4	85.1	84.8	80.6	73.2	64.8	59.2
Mean Daily Temperature Range	5% DB	MDBR	20.4	21.0	23.0	24.7	22.9	21.7	20.0	20.4	21.7	23.0	23.4	19.3
		MCDBR	25.1	25.7	28.3	29.8	26.8	25.8	22.9	24.4	26.3	26.1	27.0	22.9
		MCWBR	17.1	16.8	15.1	13.9	11.1	9.5	8.4	8.0	9.2	12.3	15.8	16.5
	5% WB	MCDBR	20.1	20.0	22.6	23.2	21.7	20.5	20.2	19.9	19.9	19.1	19.1	18.7
		MCWBR	15.3	15.5	13.4	12.1	10.1	8.9	8.5	7.6	8.1	10.4	13.7	15.2
Clear-Sky Solar Irradiance	taub		0.317	0.319	0.355	0.394	0.457	0.485	0.534	0.532	0.425	0.373	0.343	0.318
	taud		2.514	2.502	2.427	2.307	2.167	2.130	2.009	1.999	2.288	2.429	2.449	2.529
	Ebn at Noon		284	296	292	283	266	257	244	241	265	272	271	277
	Edh at Noon		26	30	35	41	48	49	56	55	39	32	28	25
All-Sky Solar Radiation	RadAvg		824	996	1347	1719	1897	1958	1866	1735	1497	1252	923	697
	RadStd		55	128	115	111	154	152	135	99	145	158	102	73

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+344	+232
Regional (40 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+274	+194

Nomenclature: See separate page